

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

192311454

I Sanjana Sree.P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sanj

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= 93
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Ques) Vehicle Fault Diagnosis System :

(a) Propositional logic Representation :

F = Engine warning light

N = Unusual noise

L = Low Oil Pressure

O = Oil leak

C = Overheating sign

F = Engine Failure Risk

S = Coolant System Failure

Rules :

1. $E \wedge N \rightarrow F$

2. $E \wedge L \rightarrow O$

3. $N \wedge C \rightarrow S$

Facts :

$E = \text{True}, N = \text{True}, C = \text{True}$

(b) Forward Chaining :

Initial facts :

E, N, C

Rule 1 :

$E \wedge N \rightarrow F$

Since E and N are true :

$$F = \text{True}$$

Rule 2 :

$$E \wedge L \rightarrow O$$

L is not given, so this rule cannot be applied.

Rule 3 :

$$N \wedge C \rightarrow S$$

Both N and C are true : $S = \text{True}$

Engine Failure Risk = True

Coolant System Failure = True

(C) Backward Chaining :

Goal : Coolant System Failure

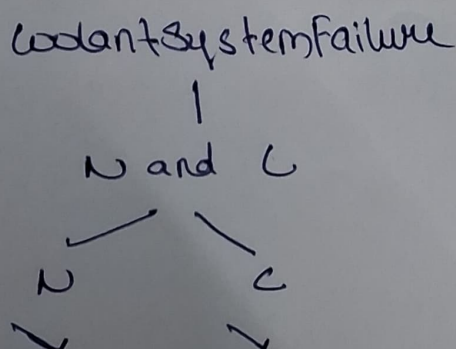
Rule 3 gives : $N \wedge C \rightarrow S$

Sub-goals are : (1) Unusual Noise (1) Overheating sign

Both are given as facts.

Coolant System Failure = True

Goal Tree :



d). Hospital Emergency ward Routing Agent :

(a) Unification of Rule 1 :

Rule 1 : $\text{Patient}(x) \wedge \text{CriticalCondition}(x) \rightarrow \text{PriorityCorridor}(x)$

Given :

Patient (Patient A)

CriticalCondition (Patient A)

Both match when :

$$\theta = \{x / \text{Patient A}\}$$

Therefore : $\text{PriorityCorridor}(\text{Patient A})$

(b) Resolution Proof for Hold (Patient B)

Rule 3 : $\text{Patient}(x) \wedge \text{waitingBehind}(x, y) \rightarrow \text{Hold}(x)$

Clf : $\neg \text{Patient}(x) \vee \neg \text{waitingBehind}(x, y) \vee \text{Hold}(x)$

Substitute : $x = \text{Patient B}$, $y = \text{Patient A}$

$\neg \text{Patient}(\text{Patient B}) \vee \neg \text{waitingBehind}(\text{Patient B}, \text{Patient A})$

$\vee \text{Hold}(\text{Patient B})$

Negated goal : $\neg \text{Hold}(\text{Patient B})$

Resolution gives : $\neg \text{Patient}(\text{Patient B}) \vee \neg \text{waitingBehind}(\text{Patient B}, \text{Patient A})$

$\text{Hold}(\text{Patient B}) = \text{True}$

(C) Handling Two critical Patients :

The KB is not completely sufficient for two simultaneous critical patients.

It can identify both as having priority corridors, but there are no rules for :

(*) Deciding which patient gets priority.

(*) Resolving conflicts

(*) Allocating the corridors.

Ques 3). Weather - Based Flight Scheduling System :

(a) CNF conversion :

P1 : StormWarning $\rightarrow$ FlightDelayNeeded

becomes :

$\neg$ StormWarning $\vee$ FlightDelayNeeded

P2 : FlightDelayNeeded $\wedge$ InternationalRoute $\rightarrow$ notifyPassengers

becomes :

$\neg$ FlightDelayNeeded $\vee$ $\neg$ InternationalRoute $\vee$

notifyPassengers

P3 : $\neg \text{NotifyPassengers} \wedge \text{InternationalRoute} \rightarrow \text{ComplianceRisk}$
becomes :

$\text{NotifyPassengers} \vee \neg \text{InternationalRoute} \leftrightarrow \text{ComplianceRisk}$

Facts : StormWarning
 $\text{InternationalRoute}$

(A) Resolution Proof from NotifyPassengers :

Goal : NotifyPassengers

Negated Goal : $\neg \text{NotifyPassengers}$

Using P2 : $\neg \text{FlightDelayNeeded} \vee \neg \text{InternationalRoute}$

Using P1 : $\neg \text{StormWarning} \vee \text{FlightDelayNeeded}$

Hence : $\text{NotifyPassengers} = \text{True}$

(C) Removing Storm Warning :

If StormWarning is removed, we cannot derive :
 FlightDelayNeeded .

P2 cannot derive : NotifyPassengers

Also we cannot assume : $\neg \text{NotifyPassengers}$

Hence,

ComplianceRisk cannot be concluded.

Ques 4) . Give Exploration Knowledge Agent

(a) Belief State and Modus Ponens :

Draft [1,3], Glow [1,1], Sparkle [2,3]

Rule 1 : Draft [1,3] $\rightarrow$ Pit adjacent to [1,3]

Possible Pit locations : [1,2] or [2,3]

Rule 2 : Glow [1,1] $\rightarrow$ LavaVent adjacent to [1,1]

Possible LavaVent locations : [1,2] or [2,1]

Rule 3 : Sparkle [x,y] $\rightarrow$ Treasure [x,y]

Therefore, Treasure [2,3]

(b) Backward chaining :

Draft [1,3] $\rightarrow$ Pit adjacent to [1,3]

Pit [1,2] or Pit [2,3]

Glow [1,1] $\rightarrow$ LavaVent adjacent to [1,1]

LavaVent [1,2] or LavaVent [2,1]

Sparkle [2,3] $\rightarrow$ Treasure [2,3]

(c) Safety of the path :

Path : [1,1] $\rightarrow$ [1,3] $\rightarrow$ [2,3]

Agent knows : Treasure [2,3]

The path is not guaranteed to be safe .